GRADUATION PROJECT

Title & Team

- Who we are — Frontend · Backend · Mobile
- Agenda: problem → what → how → demo

FRONTEND

RO
Rokaya Othman

NM
Nadine Mohsen

BACKEND

AS
Abdelrahman Sersawy

AM
Ahmed Mousa

KL
Khalid Labib

MOBILE

KQ
Khaled Qenawy

Six people · three clients · one workspace

VIRTUAL OFFICE

The Problem

- The office gave five things **free**
- Fast talk · collaboration · visible status · ownership · real interaction
- **Remote took all five**
- Five disconnected tools in their place
- *“Five tools all day, still don't know what the team is doing”*

ONSITE GAVE YOU

~~Faster communication~~
~~Better collaboration~~
~~Visible work status~~
~~Clear tasks & accountability~~
~~Real interaction~~

REMOTE →

REMOTE LEFT YOU

- × **Disconnected tools**
- × **Constant context switching**
- × **No spontaneous talk**
- × **Can't see who's available**
- × **No shared place to be**

Five tools. One job. No idea what the team is doing.

VIRTUAL OFFICE

Where Existing Tools Fall Short

- Studied them honestly — they're good at their job
- **None give you a workplace.** They give you channels
- Presence is the missing piece

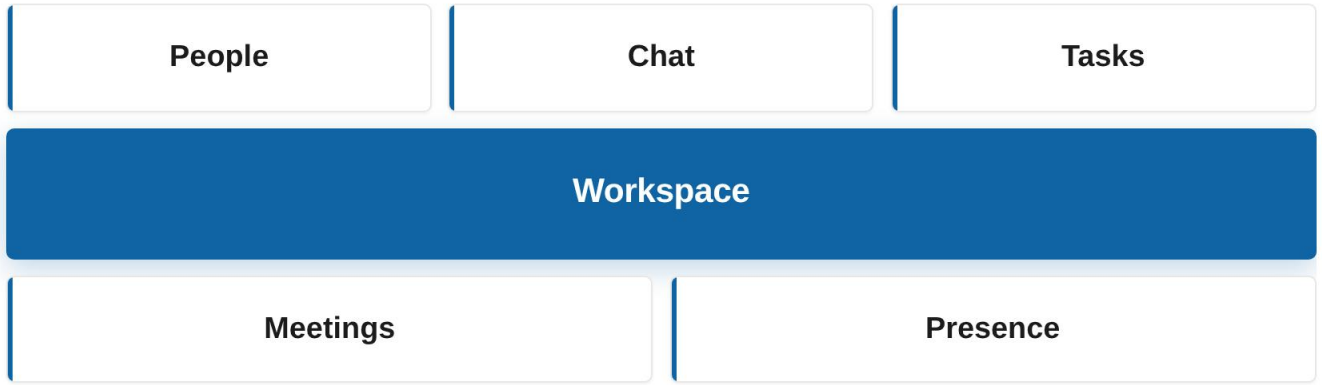
Slack · Microsoft Teams · Discord

	Slack	Teams	Discord	Ours
Chat	✓	✓	✓	✓
Threads	✓	✓	~	✓
Tasks	—	~	—	✓
Meetings	—	✓	✓	✓
Presence	~	~	✓	✓
One place	—	~	—	✓

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Our Solution: One Place

- People · Tasks · Meetings · Chat · Workspace
- Not five features — **one workflow**
- Message → task → deadline → meeting → same people



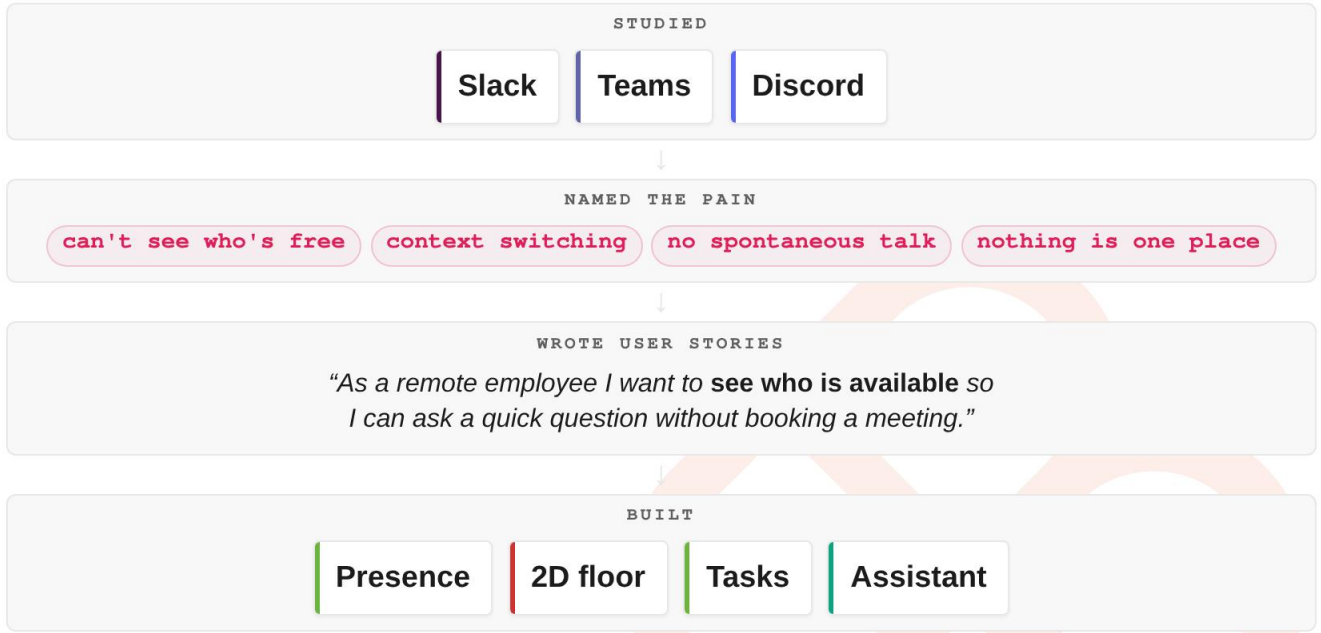
Not five features. One workflow.

VIRTUAL OFFICE

Research → User Stories

- Analysed Slack · Teams · Discord
- Studied real remote pain
- → user stories drove every feature
- *“Nothing here because it looked nice”*

Figma

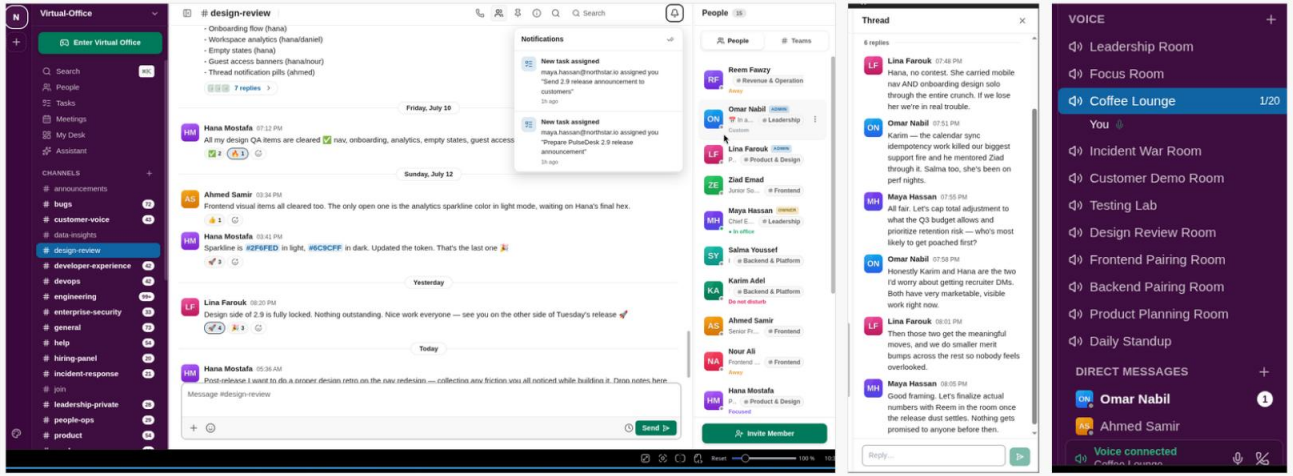


Every feature traces back to a pain. Nothing is here because it looked nice.

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Core Collaboration Features

- Chat · Threads · Voice calls
- **Parity — not the answer →**



VIRTUAL OFFICE

Catching Up Without Reading Everything

- Built for the person, not just the work
- 200 unread → nobody actually reads them
- **Ask instead of read**
- Only *your* unread — real cursor, not “last 50”
- Reading it doesn't mark it read
- Can't invent — same APIs, your permissions

Groq · Function calling

Summarise what I missed in #general today.

ASSISTANT

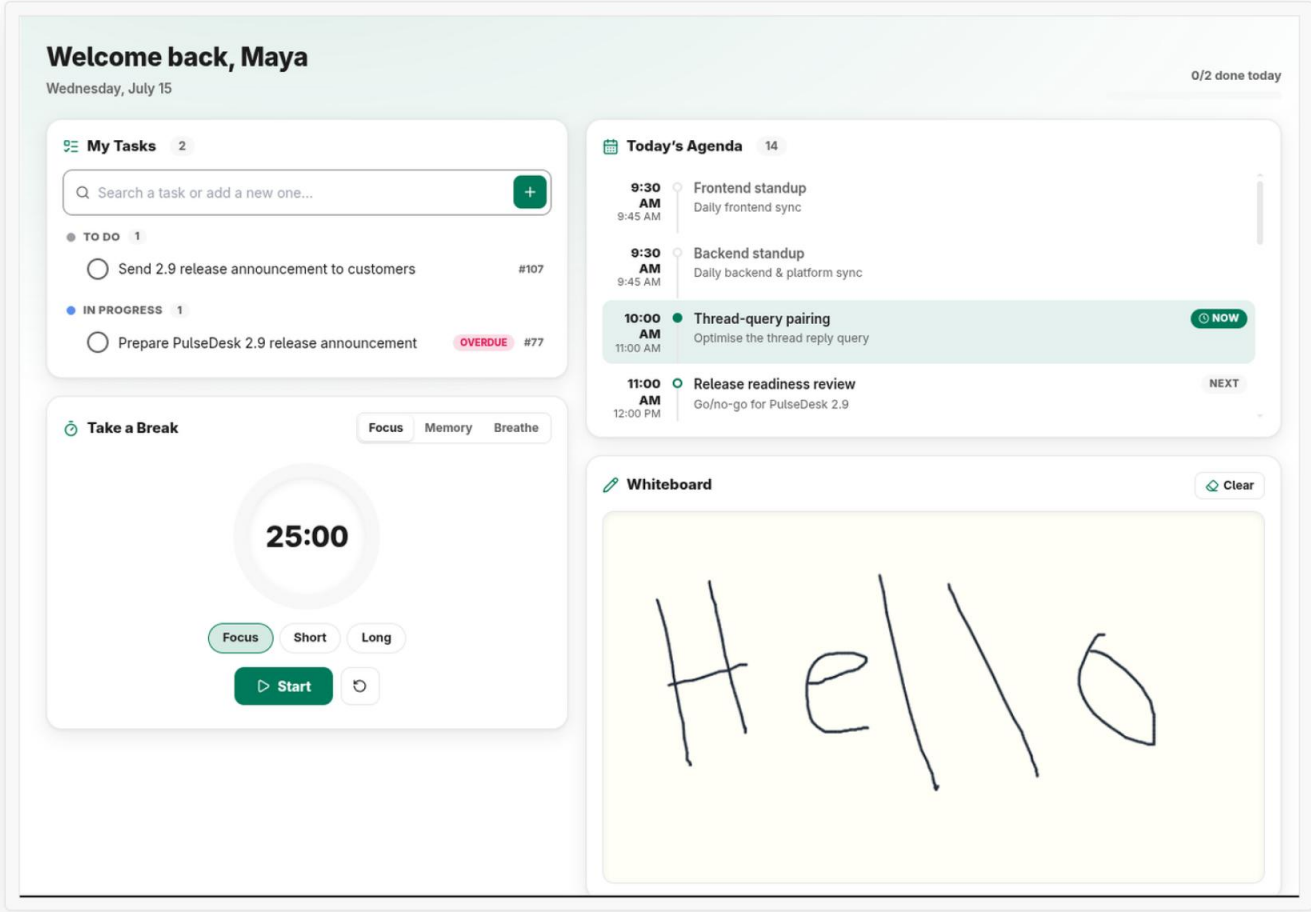
- Karim: the analytics **timezone bug is back** on staging.
- Karim asked **@maya to approve the RC cut** for tomorrow morning.
- Karim off Thursday — Salma covers the calendar-sync load test.
- Lina: launch-day runbook is pinned, everyone knows their role.

`tools used: read_channel` Real answer, real data — your cursor, your permissions.

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The Personal Desk

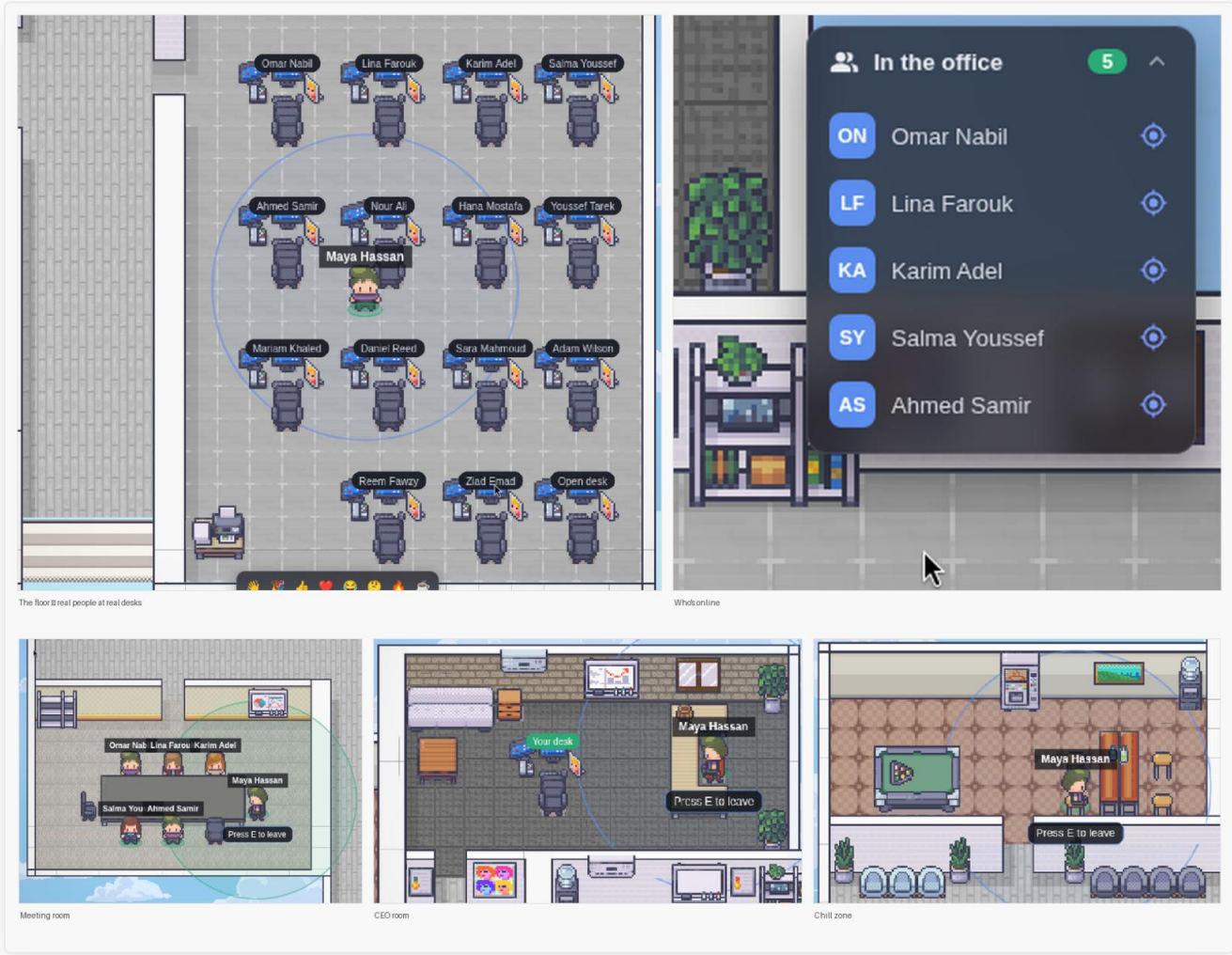
- Real office: your desk is *yours*
- Slack gives you channels. Nothing belongs to you
- My Desk — tasks · agenda · notes · whiteboard
- **Belonging is a feature**
- → *“still a room with one person in it”*



VIRTUAL OFFICE

The Biggest Gap: Human Presence

- You can't **see** your colleagues
- Real office: look up → walk over → 5 seconds
- Slack: search → DM → type → wait → maybe call
- **The 2D Plane** — see · move · walk up · join a room
- *"Not a game. Not decoration."*

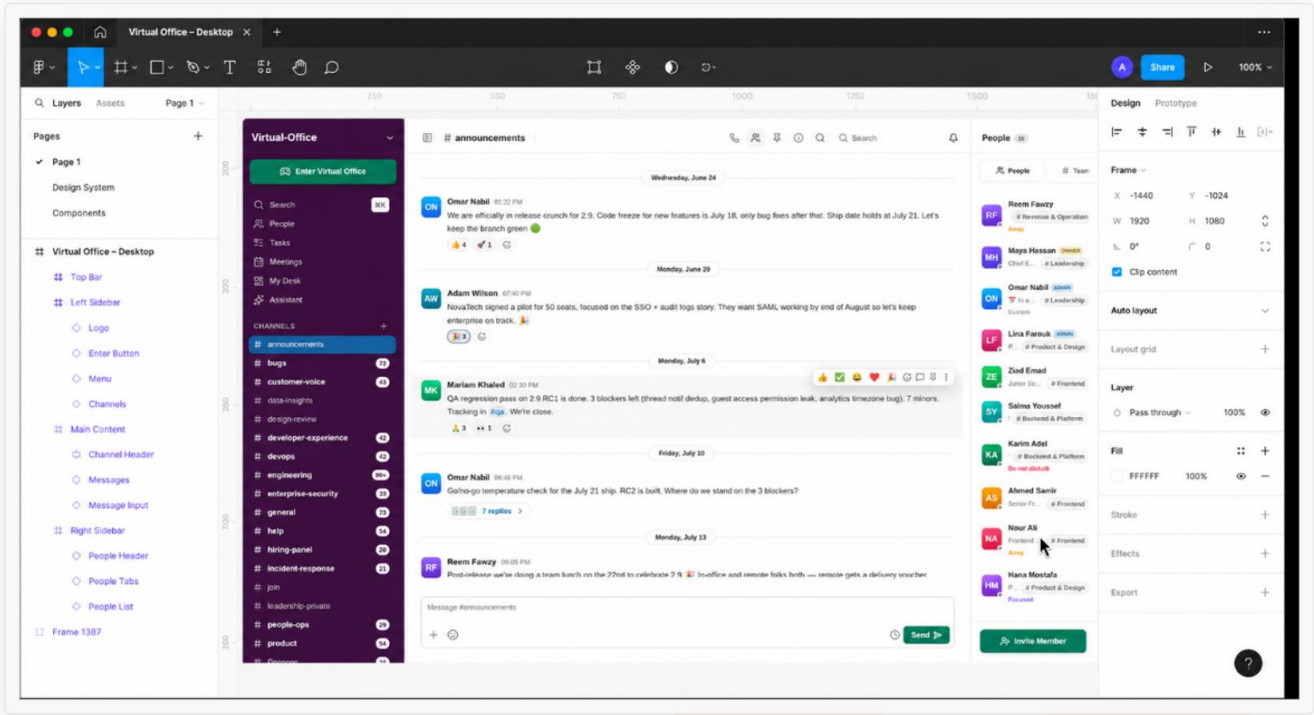


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From Requirements to Interface

- Stories → screens
- Simple · clear · reachable · consistent
- Wireframe → final

Figma



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Why React.js

- SPA — instant, no reload
- React Router v7
- Reusable: buttons · forms · modals · cards
- Axios — one place for errors + tokens
- **But HTTP: you have to ask →**

React · React Router · Axios

```
components/
├── Button.jsx
├── Modal.jsx
├── Card.jsx
└── Nav.jsx
```

built once · used on every page

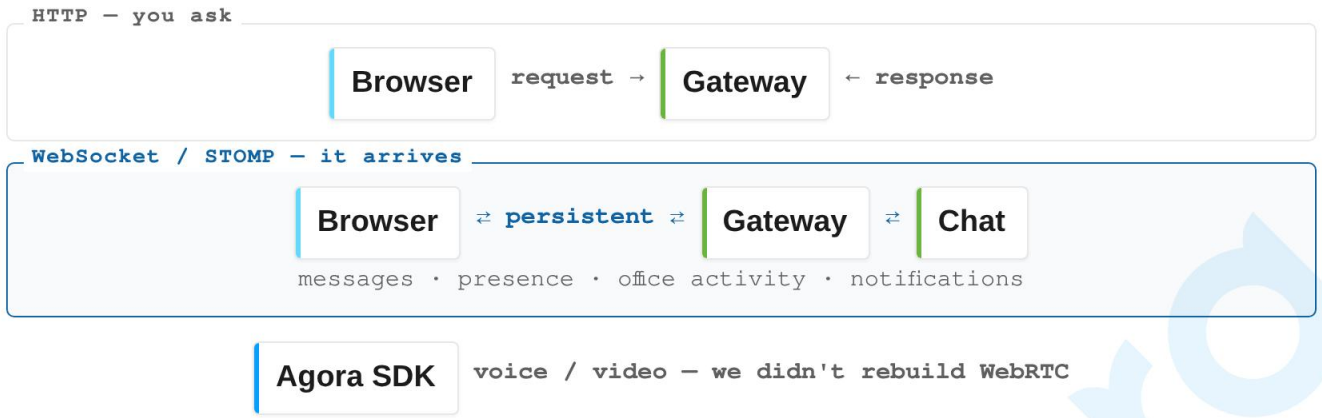
```
// one place for the token, every request
axios.interceptors.request.use(cfg => {
  cfg.headers.Authorization = `Bearer ${token}`
  return cfg
})
```

VIRTUAL OFFICE

Real-Time: WebSockets, STOMP & Agora

- HTTP you *ask* · WebSocket it *arrives*
- STOMP via `@stomp/stompjs`
- Messages · presence · office activity
- Same pipe → real-time notifications
- Agora for voice/video — we didn't rebuild WebRTC

WebSocket · STOMP · Agora

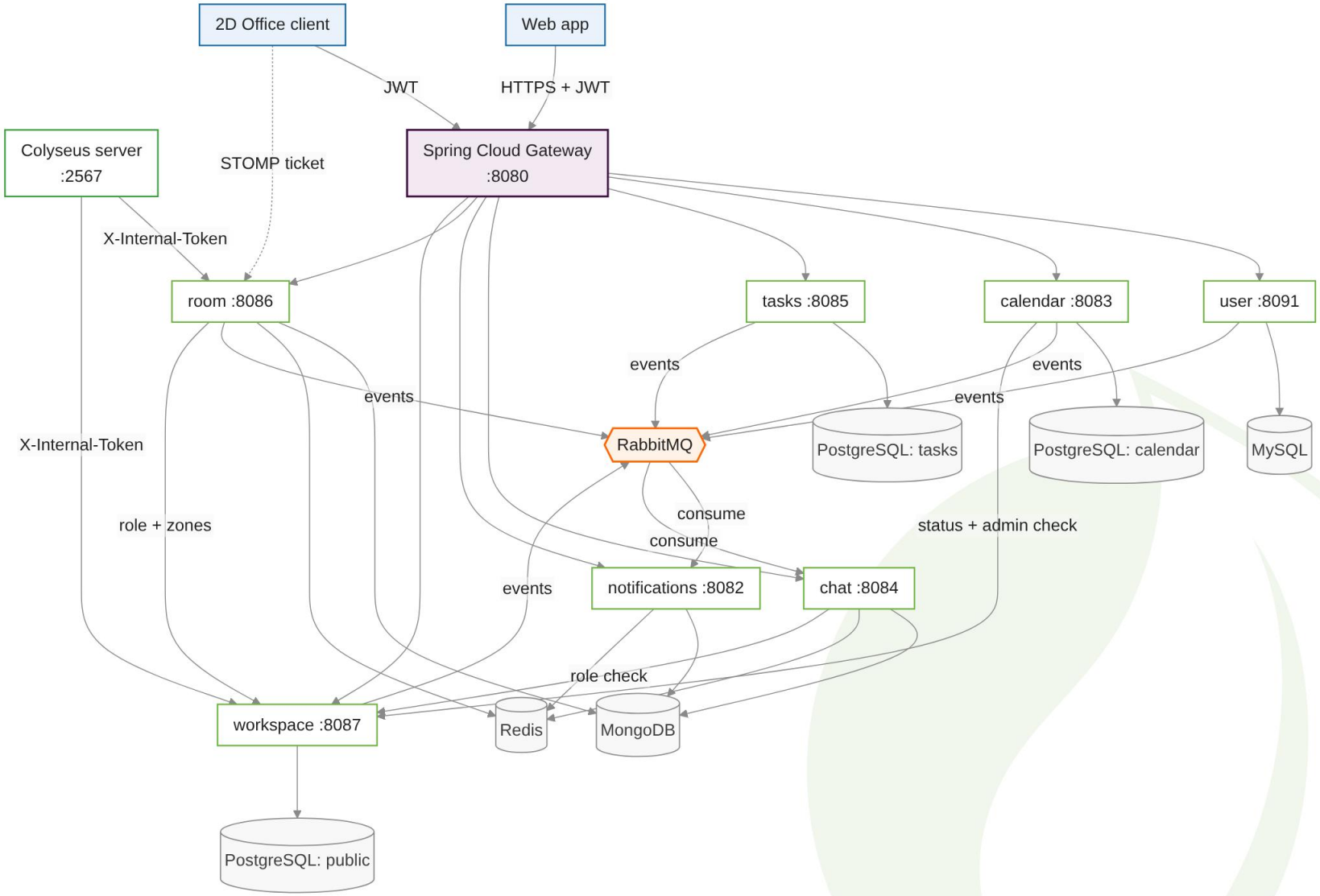


VIRTUAL OFFICE

Microservices Architecture

- Not one app — independent services
- **Gateway** — one door, validates the JWT once
- User · Workspace · Chat · Notif · Calendar · Tasks · Room · AI · +Office
- Separate · deploy alone · scale alone · **fail alone**
- ~20 containers, one command

Spring Boot · Spring Cloud Gateway · Docker · MySQL · PostgreSQL · MongoDB · Redis · RabbitMQ



Each service owns its own store. Services talk over the gateway or the broker — no shared tables, no back doors.

NEXT
MO **Mousa**
identity, tasks and the calendar

VIRTUAL OFFICE

Identity: Decided Once, at the Edge

- BCrypt — we never store a password
- JWT (HS256), shared only gateway ↔ user-service
- **Strip** client `X-User-*` · **inject** trusted
- Services never parse a JWT
- Account role **vs** workspace role
- OTP reset — no links to leak
- Register → **RabbitMQ** → welcome. Never blocks signup
- *“Decided once, at the edge”*

JWT · BCrypt · Spring Security



Services never parse a JWT. They trust the gateway — because only it can set these.

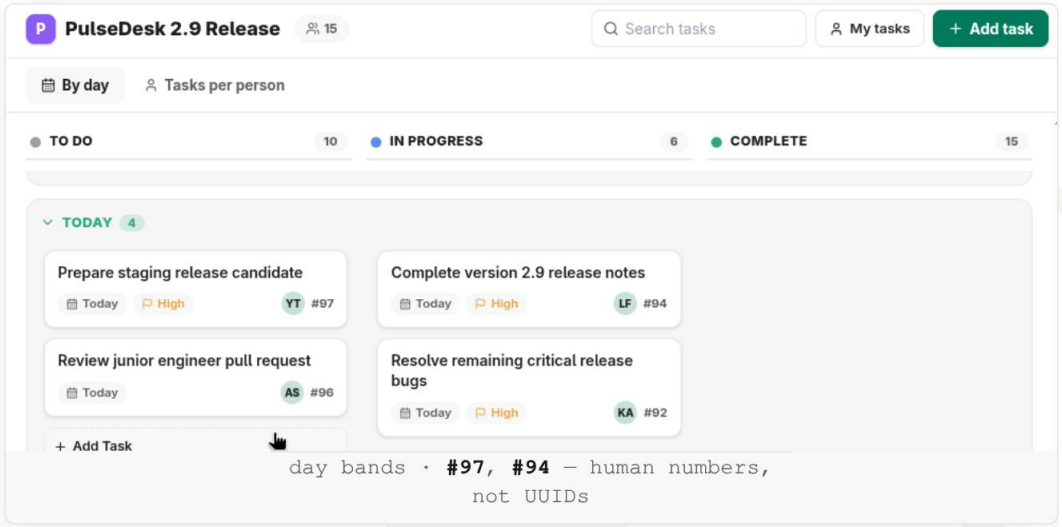
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Tasks & Calendar: The Engineering

- Spring Data JPA on PostgreSQL
- `#42`**, not a UUID** — max+1 per workspace
- **Flyway** — schema is versioned, not hand-run SQL
- Assign → RabbitMQ → Notification. **Never waits**
- `@Scheduled` 60s — reminders + **auto-status**
- Meeting starts → *"In a meeting"* by itself
- **Idempotent** — flags, never double-send
- **Never clobber a human's own status**

Spring Data JPA · PostgreSQL · Flyway · RabbitMQ

```
@Scheduled(fixedDelay = 60_000)
void sync() {
  applyForStartedMeetings(now);   // -> "In a meeting"
  revertForEndedMeetings(now);    // only if our marker
}
// fails? don't set the flag. next tick retries.
```



day bands · **#97, #94** — human numbers, not UUIDs

NEXT
KH **Khalwsh**
chat, notifications and rooms

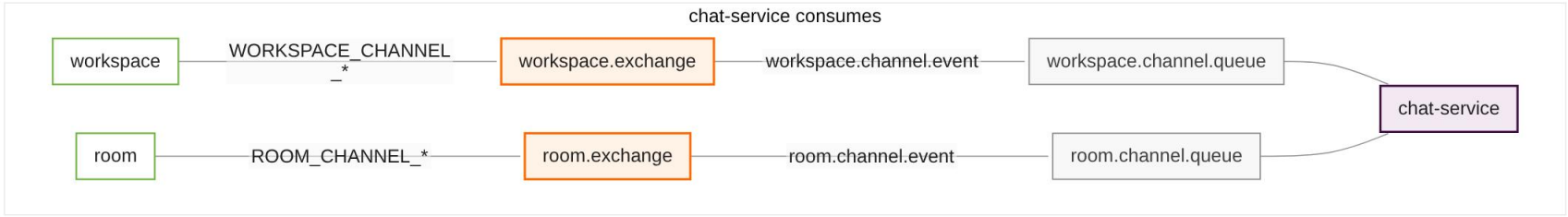
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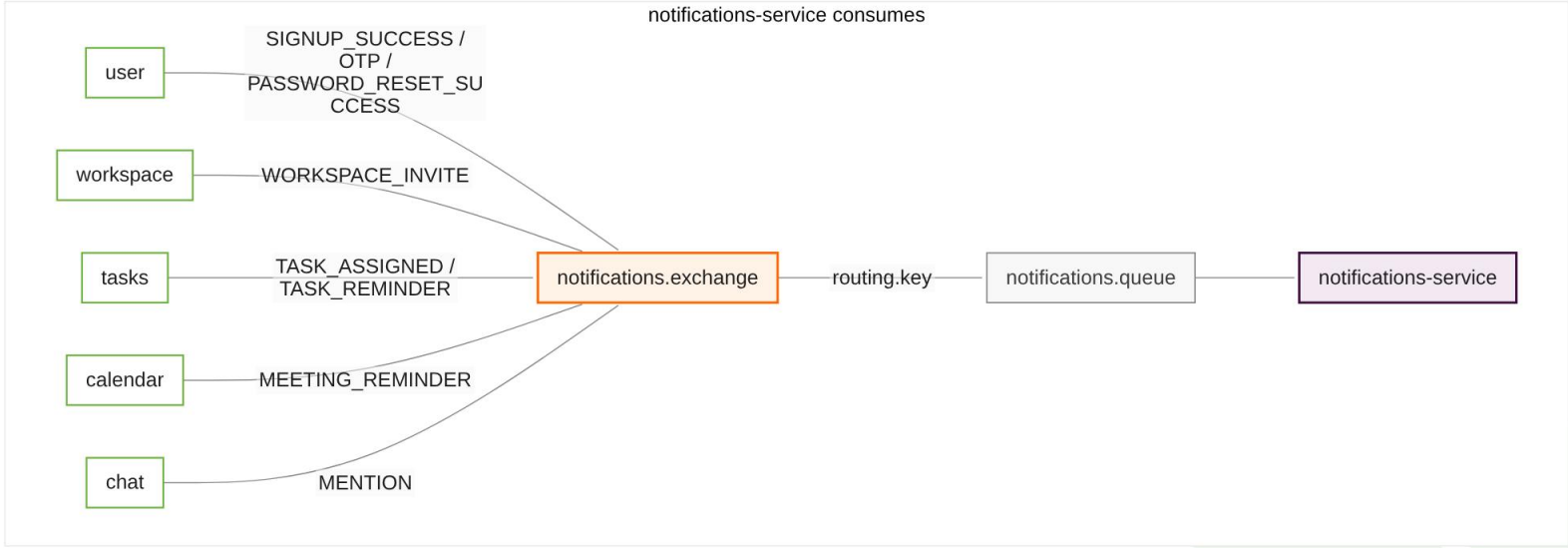
Chat, Notifications & Rooms

- Chat: DMs · channels · threads, live
- WebSocket · **Redis** presence · **Mongo** volume
- **DMs are E2E — we can't read them**
- ECDH → AES-GCM **in the browser**
- Server stores `vo1:<iv>:<ct>` and nothing else
- **Trade-off: DMs only** — channels feed search + AI
- Limits: one device · no forward secrecy · TOFU
- Notif: **RabbitMQ** — Task never waits
- Rooms: we own the permissions

MongoDB · Redis · RabbitMQ · WebSocket · WebCrypto

```
> db.messages.find({content: /^vo1:/})
vo1:Tz044O2p++/WL3yy:YHdQoK4tYMtycnZBfodXR5RX8…
```





Two consumers own the topology — notifications and chat. Every other service only produces.

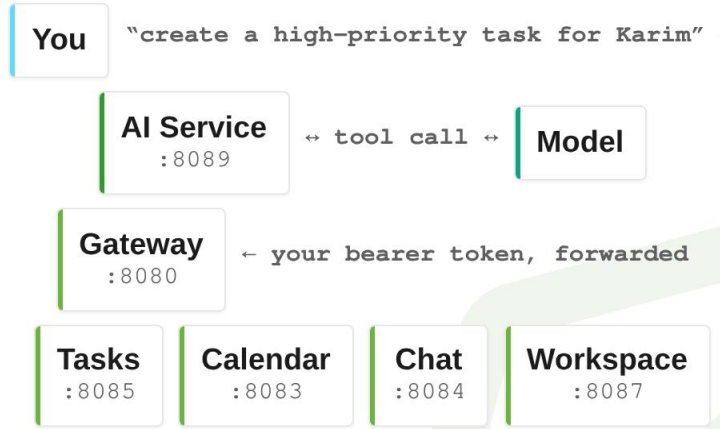
NEXT
SE **Sersawy**
the workspace and the AI bot

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Workspace Service & the AI Bot

- Workspace = the backbone
- Members · teams · channels · roles · office layout
- AI: summarise · decisions · create · answer
- **The hard part was context**
- Function calling — same APIs, **your** token

Node.js · Groq · Function calling · PostgreSQL · Flyway



No service account.

The assistant can only see and do what you could do yourself — same APIs, same permissions.

VIRTUAL OFFICE

The 2D Office: Why a Game Engine

- **Phaser 3** — a floor plan **is** a game loop
- **Colyseus** — authoritative, its own Node service
- *"Why two realtime stacks?"* → **opposite problems** →
- `@type` decorators = binary deltas = **the protocol**
- `filterBy(workspaceId)` → one room per company
- Phaser canvas + React HUD, one page
- Proximity → the call connects. You walk over
- Joystick for touch

Phaser · Colyseus · Tiled tilemap · React · Redux

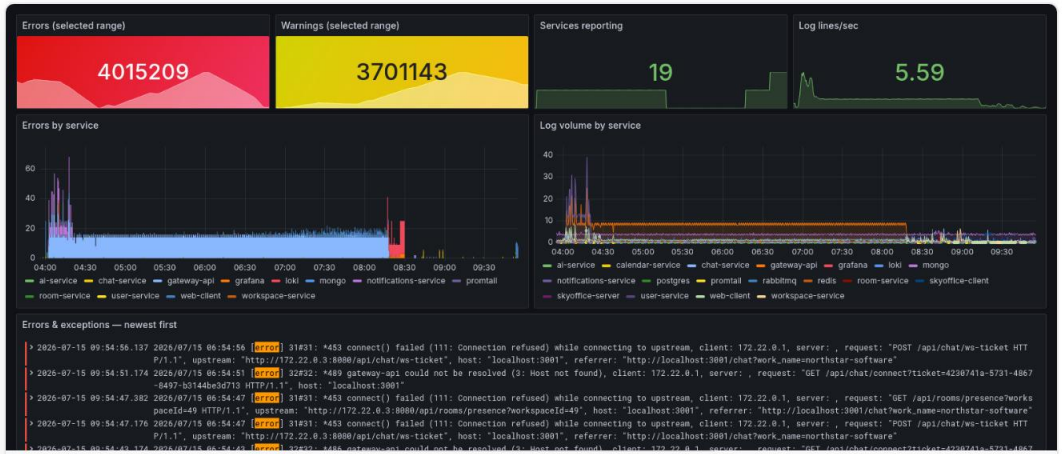
	Chat	Movement
Every event matters?	Yes — never lose one	No — a late one is garbage
Order matters?	Yes	Only the newest
Store it?	Forever (MongoDB)	Never
Rate	A few / minute	~60 / second
So	**STOMP + MongoDB**	**Colyseus**

VIRTUAL OFFICE

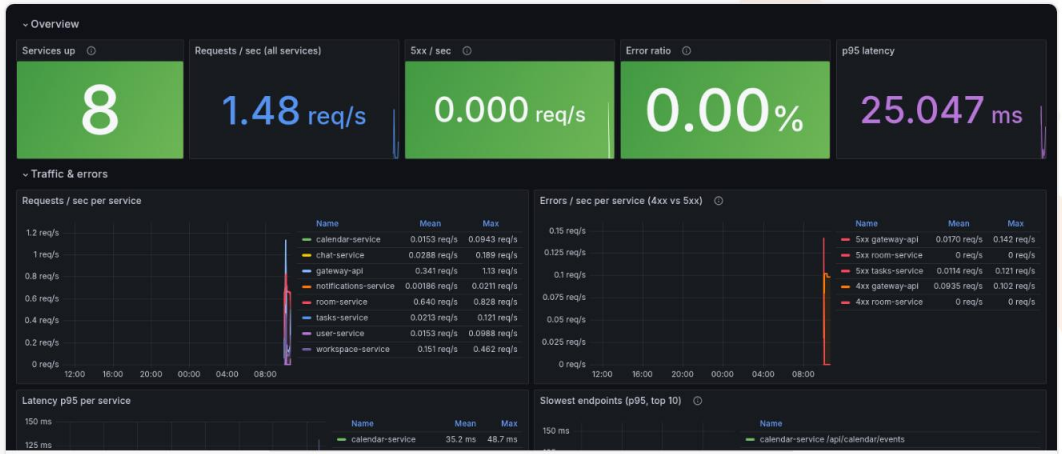
Observability: Knowing When It Breaks

- 20 containers, 9 services → nine `docker logs`?
- **Loki** ← Promtail ← every container
- One **Grafana** dashboard
- **Paid off day one:** ~200 errors/hr nobody saw
- → the WS handshake bug Khalwsh just described
- *“App looked fine. Logs didn't.”*
- **Prometheus** — 8 services scraped · p95 · error ratio
- Logs say *what*. Metrics say *how often, how slow*

Loki · Promtail · Prometheus · Grafana



Loki — logs: *what happened*



Prometheus — metrics: *how often, how slow*

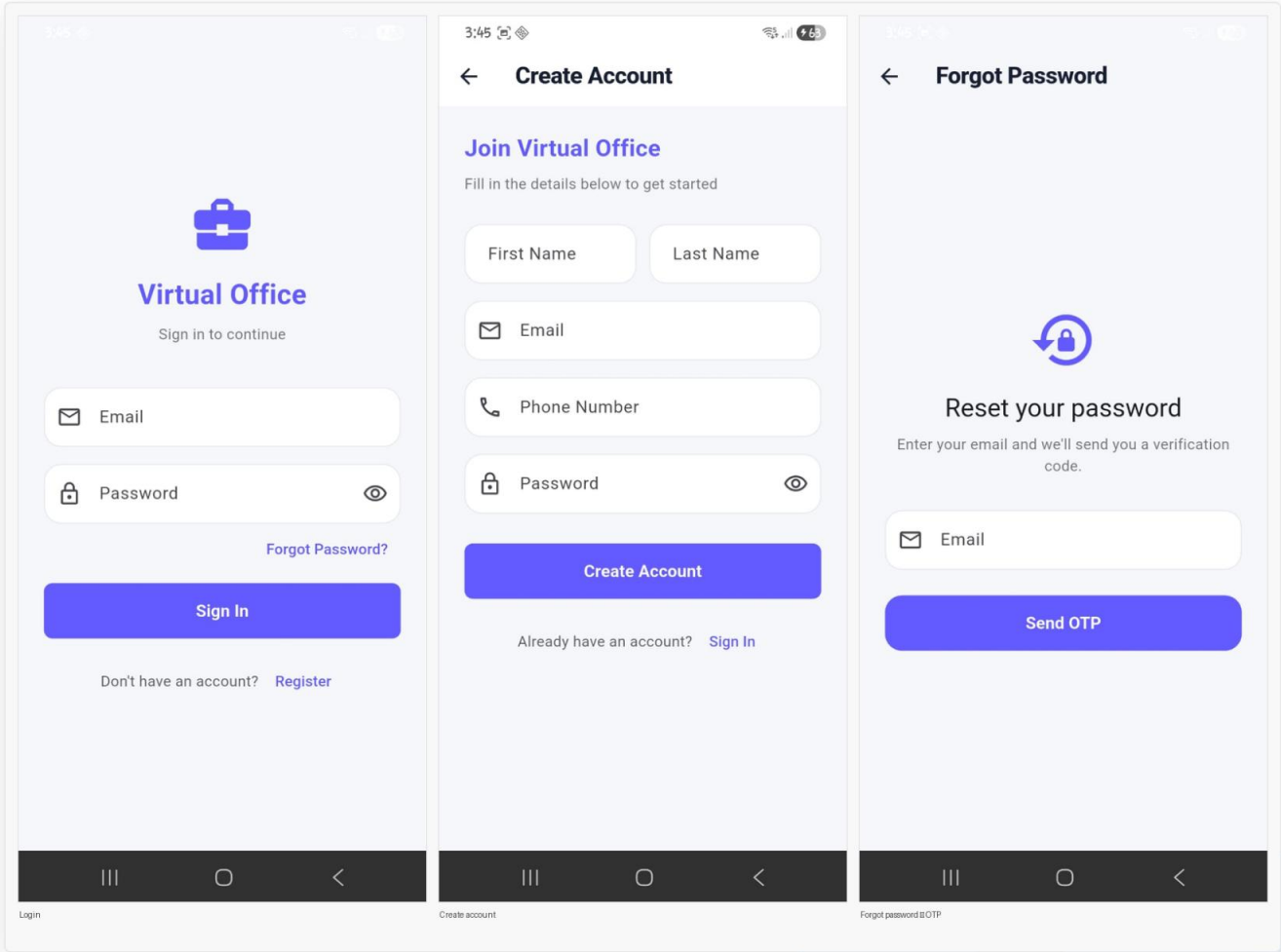
NEXT
KQ **Khaled Qenawy**
the phone

VIRTUAL OFFICE

Mobile: Why Flutter, and What We Chose to Build

- One codebase → Android · iOS · web
- **Deep on chat, not shallow on everything**
- Nobody plans a sprint on a 6-inch screen
- Tasks/calendar/calls = screens, not architecture

Flutter Dart



VIRTUAL OFFICE

Mobile: Architecture

- Feature-first: `core/` + `features/chat/`
- Screen → Provider → Repository → ApiClient
- Provider **split by concern**, not one god-object
- `shared_preferences` holds the JWT
- **Same APIs as web. No mobile backend.**

Flutter · Provider · shared_preferences

Screen → **Provider** → **Repository** → **ApiClient** → **Gateway**

```
lib/
├── core/ theme · constants
└── features/chat/
    ├── api/ · models/ · repositories/
    ├── providers/ connection · channel · message · thread
    └── screens/ · services/ · widgets/
```

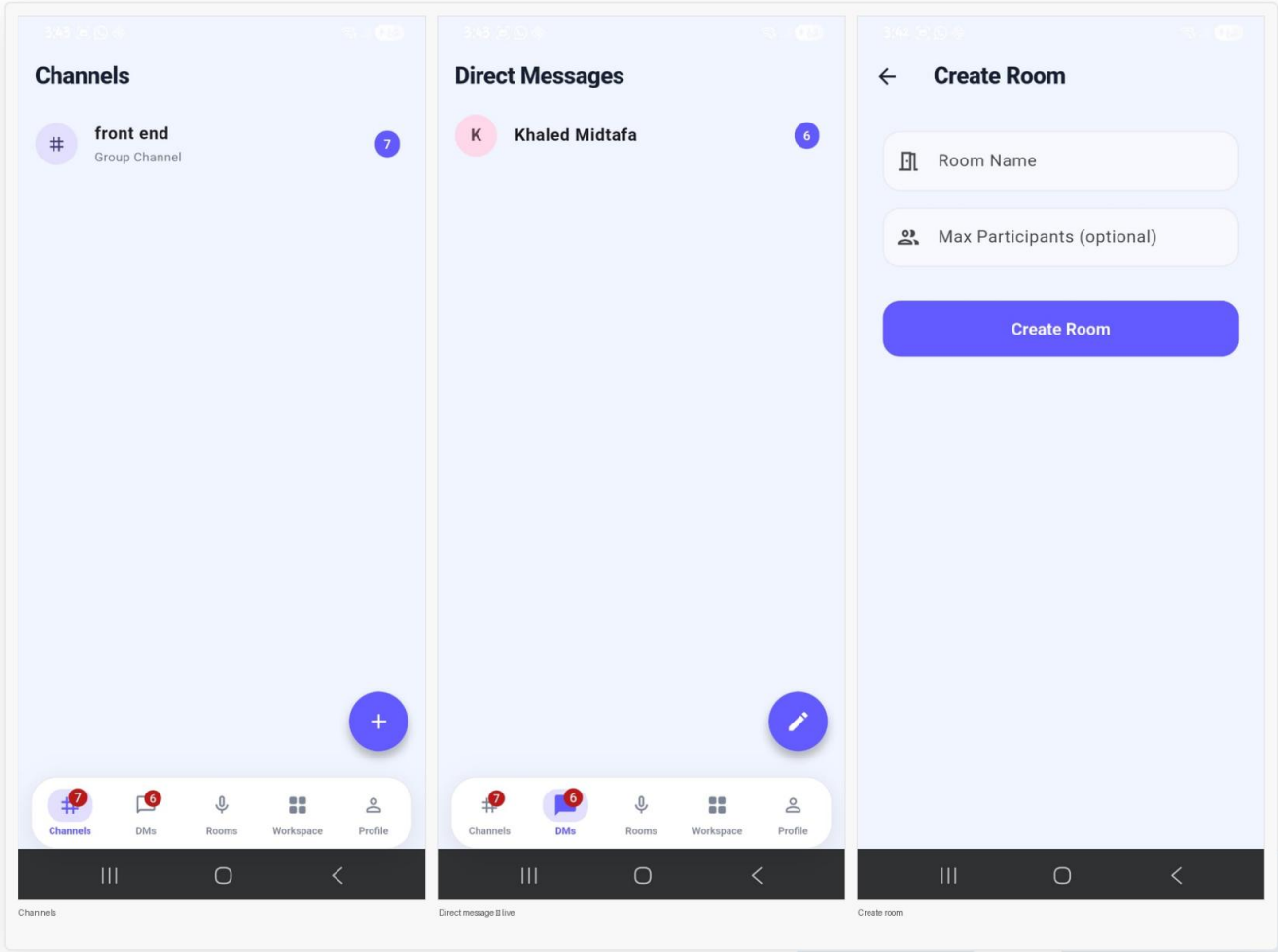
Same APIs as the web. No mobile backend.

VIRTUAL OFFICE

Mobile: Real-Time & Notifications

- **Same STOMP contract as the browser**
- `stomp_dart_client` ↔ `@stomp/stompjs`
- We reused the realtime stack, didn't rebuild it
- `uuid` → clientMessageId → idempotent retries
- `awesome_notifications` on the device
- Calls next: Agora SDK — Rooms already handles it

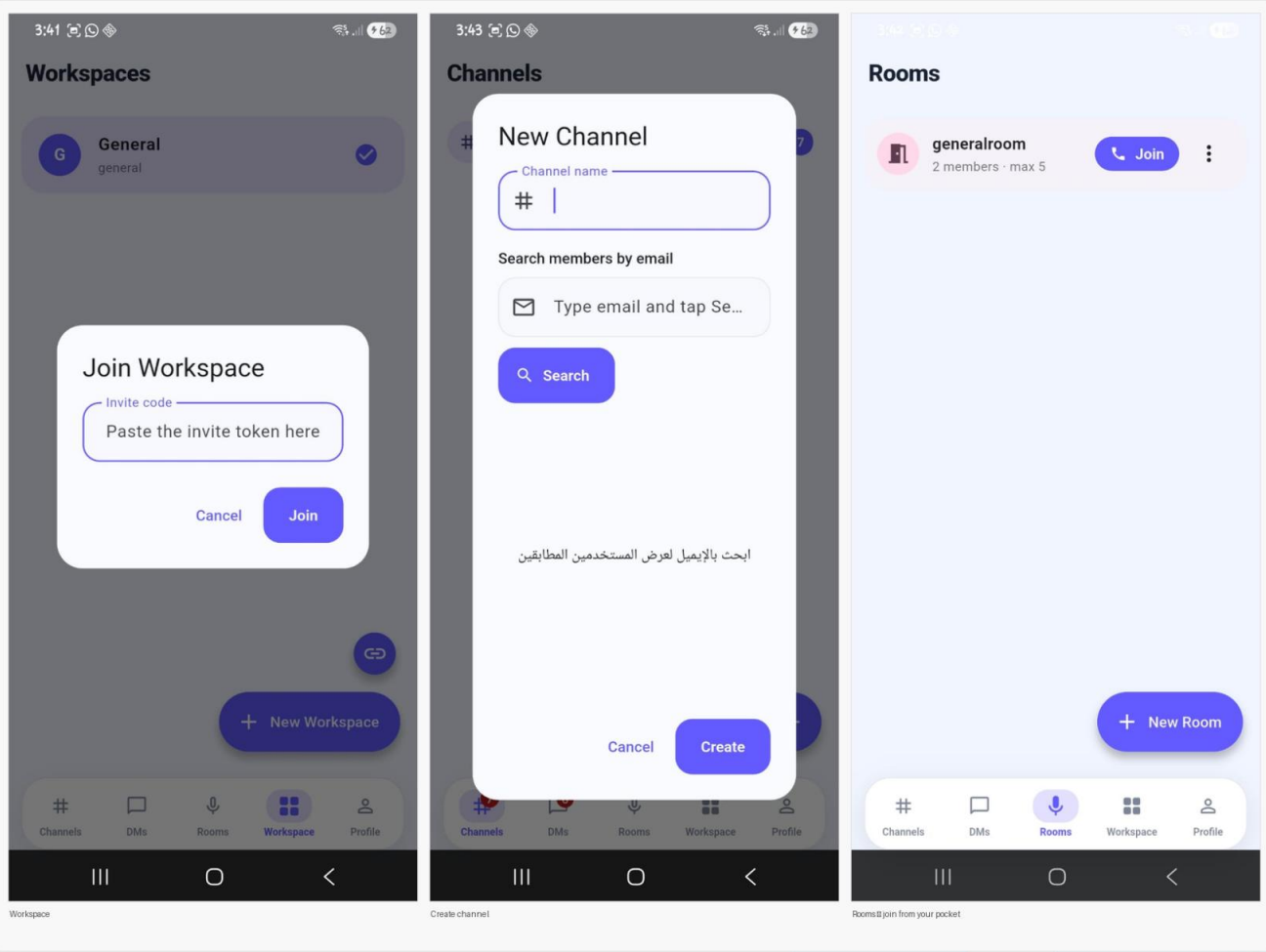
`stomp_dart_client` `awesome_notifications`



VIRTUAL OFFICE

Mobile: The Whole Workspace, in a Pocket

- Five tabs: Channels · DMs · **Rooms** · Workspace · Profile
- Not a chat window — **the workspace**
- Create a channel on the phone → it's in the browser's list
- Join a voice room from your pocket
- **No mobile backend** — same gateway, same JWT, same permissions



VIRTUAL OFFICE

Live Demo & Close

- Login → the workspace is **alive**
- Chat + thread
- *“Catch me up”*
- Create a task from a sentence
- **Walk into the 2D office**
- *“…one place you can actually walk into.”*

1. **Log in** — the workspace is already alive. 15 people, real history.
2. **Chat + thread** — open #general, open a thread, send one message.
3. **“Catch me up”** — the assistant answers from the real cursor.
4. **Create a task from a sentence** — watch it land on the board.
5. **Walk into the 2D office** — end here. Always end here.

Backup video ready. If a step stalls, move on — never debug on stage.